



Review or Mini-review

Association of maternal arsenic exposure with birth size: A systematic review and meta-analysis

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ABSTRACT

Objectives: Chronic exposure to arsenic during pregnancy or early life continues to be a major global health problem worldwide, affecting hundreds of millions of people. However, its effects on birth size are uncertain. The aim of this study is to summarize the association between maternal arsenic exposure and birth size.

Methods: A comprehensive literature search was conducted on the maternal arsenic exposure and birth size (birth weight, birth length, head circumference and chest circumference) through databases (PubMed, Qvid Medline, Web of Science, Cochrane and EMBASE), the last search was in March 2019. Heterogeneity was tested by using the Q statistic and stratifying for epidemiological factors. The possibility of publication bias was assessed through Begg's test.

Results: A total of 12 studies provided sufficient data were included. Meta-analysis revealed that maternal arsenic exposure was associated with the decrease of birth weight ($\beta = -25.0$ g; 95% CI: $-41.0, -9.0$) and head circumference ($\beta = -0.12$ cm; 95% CI: $-0.24, -0.01$) in random-effect model, birth length ($\beta = -0.12$ cm; 95% CI: $-0.17, -0.07$) in fixed-effect model, no significant reduction was suggested in chest circumference ($\beta = -0.74$ cm; 95% CI: $-1.78, 0.3$). In addition, we found that the birth weight decreased stronger in women living in the Americas, or applying the exposure marker of urine.

Conclusions: Maternal arsenic exposure is associated with the decrease of the birth weight, length and head circumference. Millions of people are still exposed to high-level arsenic, therefore, high quality epidemiologic studies that involve large samples are necessary to determine the precise relationships between maternal arsenic exposure with birth size. And that effective mitigation measures are also need to prevent arsenic exposure in women of reproductive age.

1. Introduction

Arsenic exposure through drinking water, air, food is an environmental public health problem worldwide. More than 120 million people are affected, major groups live in developing countries, such as Bangladesh, China and India (Milton et al., 2017). And drinking water plays an important role in arsenic exposure of human, maybe due to the higher value in natural background or the filter degradation of membrane bioreactor caused by severe biofouling (Sepehri and Sarrafzadeh, 2018). The World Health Organization had set the guideline level of 10 µg/L for arsenic in drinking water. But there are still about 100 million population living in Bangladesh, China and India exposure to arsenic levels more than 50 µg/L. It is estimated that 46% of the

residents in Bangladesh is exposure to arsenic concentrations above 10 µg/L (Kile et al., 2016). And the range of urine arsenic for 1578 Bangladesh women was 6–978 µg/L (Rahman et al., 2009). The Northern Finland Birth Cohort program showed that participants' blood concentration of arsenic was 0.47 ± 2.78 µg/L (Abass et al., 2016). Meanwhile, the concentration of blood arsenic in China women was 0–3590 µg/L (Guan et al., 2012). Therefore, potential health risk of arsenic exposure should be addressed, especially in women of reproductive age.

The adverse health outcomes of exposure to arsenic which depend on dose, frequency and duration have been extensively explored from children to adults (Navas-Acien and Nachman, 2013). Epidemiologic studies have shown that chronic exposure to arsenic is associated with a

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number of adverse human health effects, range from neurological and respiratory damage (Smith et al., 1998; Hertz-Picciotto et al., 2000). Meanwhile, there is increasing concern on the association between maternal arsenic exposure in early life, pregnancy and child development adversely (Concha et al., 1998). Arsenic can cross the placenta readily, some studies revealed that exposure to arsenic during pregnancy might increase the risk of fetal loss, infant morbidity and mortality and lower birth size (Rahman et al., 2010).

Birth size, a reflection of intrauterine growth and development, which was a crucial factor of morbidity and mortality in childhood, and evoking concerns for chronic diseases in adulthood (Kajantie et al., 2005). The association about maternal arsenic exposure and birth size is an area of increasing research interest. Several previous studies had examined the relation between maternal arsenic exposure and birth size, but the findings are equivocal (Rahman et al., 2009; Kippler et al., 2012; Davis et al., 2015; Bloom et al., 2015). Given that studies on the association of maternal arsenic exposure with birth size are fragmented and less roundly, and that the results are also inconsistent. Therefore, the objective of our meta-analysis included 12 studies to pool the estimation between maternal arsenic exposure on birth weight, birth length, head circumference and chest circumference, respectively. In addition, we explored related factors that influence maternal arsenic exposure for birth size.

2. Material and methods

2.1. Search strategy and selection criteria

The database of PubMed, Qvid Medline, Web of Science, Cochrane and EMBASE (up to March 2019) were searched to obtain studies or relevant publications describing the association between arsenic exposure and birth size (Qin et al., 2013). The keywords applied were “arsenic”, “arsenicals”, “arsenate”, or “arsenite” and “birth size”, “birth weight”, “head circumference”, “birth length”, “chest circumference”, “pregnancy outcome”, “birth outcome”, “reproductive outcomes” or “newborn outcomes”. We also conducted a hand search of bibliographies of relevant studies and recent review articles.

Two investigators independently evaluated the eligibility records according inclusion and exclusion criteria. The literature search and review process were conducted in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) criteria and Meta-Analysis of Observational Studies in Epidemiology (MOOSE) guidelines (Moher et al., 2009). Studies that meeting the following eligibility criteria: (1) original study; (2) human study; (3) English language; (4) arsenic exposure with one or more of the following outcomes: birth weight, birth length, head circumference, and chest circumference. Studies were excluded if it was an ecological design or one reviews.

2.2. Data extraction and quality assessment

The following information of the included studies were extracted: the name of first author, publication year, country, the number of participants, marker of arsenic exposure (urine, water, hair and blood), dose of exposure, related data and classification on birth size. If the data provided in the original article was incomplete, we contacted the author by email for detailed information. We assigned the range, mean or median of arsenic exposure from each study as the exposure level for all participants in that category. All measures of association between maternal arsenic exposure and birth size were pooled together as approximation of the relative risk. If the confidence intervals or standard errors of regression coefficient were not reported, we conservatively approximated the standard error using the reported P-values and sample size (Moon et al., 2017).

2.3. Statistical analysis

The pooled regression coefficient (β) of included studies and its corresponding 95% confidence intervals (CI) were assessed by the forest plots. The heterogeneity between various studies was mainly evaluated by Q statistic. And the effect of heterogeneity in each study was also estimated by the method ($I^2 = [(Q - df)/Q] \times 100\%$, ranges from 0% to 100% with lower values representing less heterogeneity). Before analysis, we controlled the regression coefficient of birth weight divided by the -100, the regression coefficient of birth length, head circumference and chest circumference divided by the -1, then pooled the transformed regression coefficient using the random-effect model if the values of $I^2 > 25\%$ to represent large heterogeneity, otherwise using the fixed-effects models (Quansah et al., 2015). The result multiplied the corresponding transformational multiple in the last. The Egger's test was used to explore the publication bias of our meta-analysis. We also conducted a subgroup analysis for birth size according to the location, marker of exposure and study design. The analysis were performed with Stata 12.0 software (StataCorp LP, College Station, TX).

3. Results

3.1. Study characteristics

Initially, a total of 2989 studies were identified, and forty-two studies were included after screening their titles, abstracts and whole article. Thirty-two studies were excluded for various reasons included incompatible exposure definition ($n = 9$), incompatible outcome definition ($n = 13$), application of job title ($n = 7$) and duplicated or overlap datasets ($n = 3$). Additional two studies identified by reading the reference of included articles, twelve studies were selected in our meta-analysis at last (Fig. 1) (Hopenhayn et al., 2003; Yang et al., 2003; Huyck et al., 2007; Rahman et al., 2009; Myers et al., 2009; Guan et al., 2012; Laine et al., 2015; Bloom et al., 2015; Kile et al., 2016; Henn et al., 2016; Liu et al., 2017; Liao et al., 2018), including 32,457 participants. Among these included articles, twelve studies focused on birth weight, five of which also included results for birth length and six for head circumference, three included results for chest circumference. Table 1 summarizes the characteristics of the included studies, sources of arsenic exposure assessments, dose of exposure, and the type of adverse pregnancy. Of the 12 studies, most were done in China ($n = 5$), Bangladesh ($n = 3$) and the Americas ($n = 4$) included the USA ($n = 3$) and the Chile ($n = 1$). Of these, eight studies applied biomarkers in urine ($n = 5$), hair ($n = 1$) and blood ($n = 2$), and the remaining studies measured arsenic dose in the drinking water ($n = 4$). Most studies (11/12) provided the dose of maternal arsenic exposure, and ten studies were prospective cohort study with high quality (Table 1). The adjusted

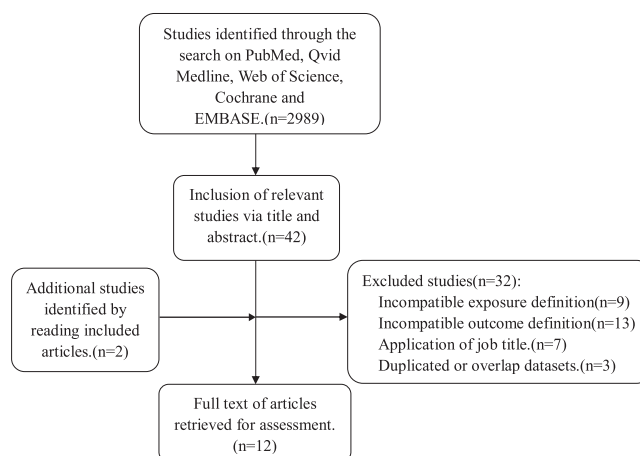


Fig. 1. Flow diagram illustrating the process of study identification.

Table 1

Characteristics of studies included in the systematic review and meta-analysis.

Sources	Location	Study population	Arsenic concentration			Outcome studied	Study design
			Exposure marker	Exposure contrast	Range/median/mean		
Hopenhayn et al. (2003)	Chile	844 women	Water	40 µg/L vs. < 1 µg/L	32.9–52.7 µg/L	BW	PCO
Yang et al. (2003)	China	18259 women	Water	NA	0–3590 µg/L	BW	PCO
Huyck et al. (2007)	Bangladesh	49 women	Hair	≥ 2.70 µg/g vs. < 0.28 µg/g	0.14–3.28 µg/g	BW	PCO
Rahman et al. (2009)	Bangladesh	1578 women	Urine	≥ 100 µg/L vs. < 100 µg/L	6–978 µg/L	BW,HC,BCC	PCO
Myers et al. (2009)	China	6905 women	Water	> 100 µg/L vs. < 20 µg/L	Not reported	BW	CS
Guan et al. (2012)	China	125 women	Blood	NA	Median = 5.30 µg/L	BW,BL,HC,BCC	CS
Laine et al. (2015)	USA	200 women	Urine	NA	1.9–448.2 µg/L	BW,BL,HC,	PCO
Bloom et al. (2015)	USA	215 women	Urine	NA	17.13 µg/L	BW, BL, HC	PCO
Kile et al. (2016)	Bangladesh	1140 women	Water	NA	Median = 2.3 µg/L	BW	PCO
Henn et al. (2016)	USA	622 women	Blood	IQR increase	1.0–2.3 µg/L	BW, HC	PCO
Liu et al. (2017)	China	1390 women	Urine	NA	1.9–448.2 µg/L	BW,BL	PCO
Liao et al. (2018)	China	130 women	Urine	NA	GM = 40.0 µg/L	BW,BL,HC,BCC	PCO

Abbreviations: NA not applicable; BW birth weight; BHC birth head circumference; BL birth length; BCC birth chest circumference; GM geometric means; CS cross-sectional study; PCO prospective cohort study; IQR interquartile range.

variable in the included studies mainly were gestational age, sex of infant, maternal education and tobacco smoke.

3.2. Arsenic exposures and pregnancy outcomes

3.2.1. Birth weight

In all, twelve studies reported regression coefficient with standard errors and provided data for our quantitative analysis. Seven studies observed a decrease in birth weight in women with highest concentrations of arsenic and four did not observe a statistically significant association. Meanwhile, Myers et al reported that the newborn birth weight of 6905 women exposed to high arsenic dose (> 100 µg/L) in water was increased ($\beta = 50$ g, 95% CI: 20.0, 80.0). The summary regression coefficient in populations exposed to arsenic showed a decreased association for the birth weight (−25.0 g; 95% CI: −41.0, −9.0; p-value for heterogeneity = 0.000; $I^2 = 73.3\%$; Fig. 2) with the random-effect model. Our meta-analysis suggests that maternal arsenic exposure during early pregnancy is inversely associated with newborns'

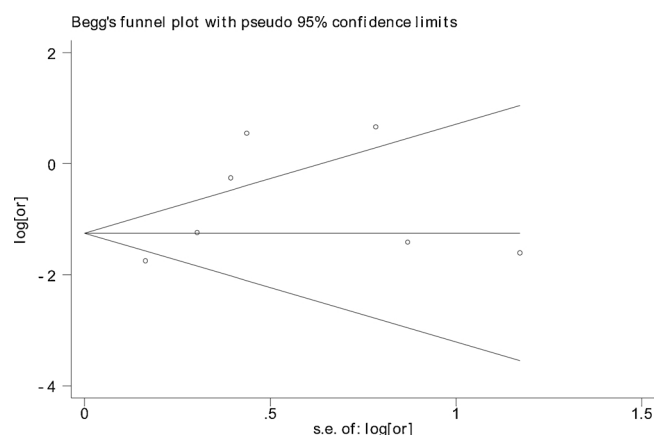


Fig. 3. The funnel plots of maternal arsenic exposure and birth weight.

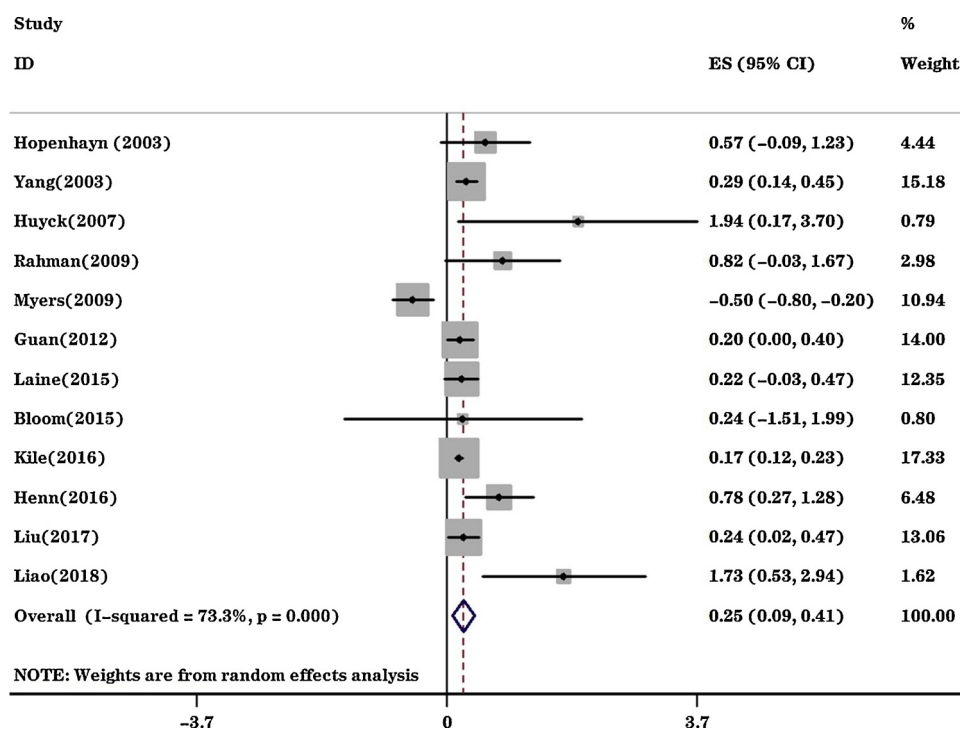


Fig. 2. Forest plot for the relation between arsenic exposure and birth weight.

Table 2
Subgroup analysis of arsenic and birth weight.

Stratification group	n	β (95%CI)	Heterogeneity test			Publication bias	
			Q	p	I^2 (%)	t	p
Total	12	-25.0(-41.0, -9.0)	41.23	0.000	73.3	1.71	0.147
Location							
China	5	-16.3(-47.2,14.5)	28.42	0.000	85.9	-0.06	0.957
Bangladesh	3	-64.5(-143.7,14.8)	4.51	0.72	0	/	/
Americas	4	-42.7(-74.1, -11.2)	4.27	0.234	29.7	/	/
Combined	12	-25.0(-41.0, -9.0)	41.23	0.000	73.3	1.71	0.147
Exposure marker							
Water	4	-9.2(-35.7,17.3)	22.91	0.000	86.9	/	/
Urine	5	-37.7(-68.0,-7.4)	7.53	0.110	46.9	/	/
Hair	1	-19.35(-369.9,-17.1)	/	/	/	/	/
Blood	2	-43.9(-99.4,11.6)	4.37	0.037	77.1	/	/
Combined	12	-25.0(-41.0, -9.0)	41.23	0.000	73.3	1.71	0.147
Marker source							
Human body	8	-40.8(-64.2,-17.5)	15.27	0.033	54.1	-1.02	0.381
Environment	4	-9.2(-35.7,17.3)	22.91	0.000	86.9	/	/
Combined	10	-25.0(-41.0, -9.0)	41.23	0.000	73.3	1.44	0.222
Study design							
PCO	10	-33.1(-48.1,-18.1)	20.63	0.014	56.4	2.23	0.09
CS	2	14.0(-54.5,82.6)	14.66	0.000	93.2	/	/
Combined	12	-25.0(-41.0, -9.0)	41.23	0.000	73.3	1.71	0.147

Abbreviations: CS, cross-sectional study; PCO, prospective cohort study; /, no value.

birth weight. The funnel plots of maternal arsenic exposure with birth weight suggested that there was no publication bias ($t = 1.71$, $P = 0.147$; Fig. 3).

In order to further explore the contribution of study variables (such as location, exposure marker, and study design) for the significant heterogeneity of this study, we conducted subgroup analysis. Strong heterogeneity was observed in our study (p-value for heterogeneity = 0.000; $I^2 = 73.3\%$), which was mainly attributed to the studies conducted in China (p-value for heterogeneity = 0.000; $I^2 = 85.9\%$), water as the exposure marker of arsenic (p-value for heterogeneity = 0.000; $I^2 = 86.9\%$) and the cross-sectional study (p-value for heterogeneity = 0.000; $I^2 = 93.2\%$) (Table 2). Furthermore, the effect of high maternal arsenic exposure for newborn birth weight was elevated in population living in the Americas ($\beta = -42.0$ g; 95% CI: -74.1, -11.2; p-value for heterogeneity = 0.234; $I^2 = 29.7\%$), and the source of exposure marker from human body ($\beta = -40.8$ g; 95% CI: -64.2, -17.5; p-value for heterogeneity = 0.033; $I^2 = 54.1\%$) compared with other populations (Details shown in Table 2).

3.2.2. Birth length

Of the five studies investigated the relation between maternal arsenic exposure and birth length, most of them were prospective cohort study. And three studies observed a decrease in newborn birth length. Summary regression coefficient indicated a reduction of birth length ($\beta = -0.12$ cm; 95% CI: -0.17, -0.07; p-value for heterogeneity = 0.917; $I^2 = 0.0\%$) (Fig. 4). In subgroup analysis, the risk of birth length was decreased more in studies applying biomarkers of urine ($\beta = -0.14$ cm; 95% CI: -0.21, -0.06), and population in the Americas ($\beta = -0.17$ cm; 95% CI: -0.31, -0.07) (Table 3). The heterogeneity was also mainly in the studies conducted in the Americas (p-value for heterogeneity = 0.048; $I^2 = 67.0\%$). Egger's linear regression test was applied to assess asymmetry of significant publication bias, and the result showed that there was no publication bias between maternal arsenic exposure and newborns' birth length ($t = 0.96$, $P = 0.512$) (Table 3).

3.2.3. Head circumference

In total of six studies examined the association between arsenic and birth head circumference. Four and two studies were conducted in women exposed to high arsenic by applying biomarkers of urine and

blood, respectively. The overall summary regression coefficient was ($\beta = -0.12$ cm; 95% CI: -0.24, -0.01; p-value for heterogeneity = 0.030; $I^2 = 59.6\%$) through the random-effect model (Fig. 5). The decrease of regression coefficient was high in subgroup of population living in the Bangladesh ($\beta = -4.07$ cm; 95% CI: -0.38, -7.75; p-value for heterogeneity = 0.02; $I^2 = 81.5\%$), and the prospective cohort study ($\beta = -0.21$ cm; 95% CI: -0.46, -0.03; p-value for heterogeneity = 0.020; $I^2 = 65.6\%$) (Table 4). And that, strong heterogeneity was also mainly attributed to these two group. There was no evidence of publication bias between maternal arsenic exposure and newborns' birth head circumference ($t = -0.41$, $P = 0.719$) (Table 4).

3.2.4. Chest circumference

A total of three studies were examined to explore the association between maternal arsenic exposure and birth chest circumference. The pooled effect did not show a significant reduction in birth chest circumference ($\beta = -0.74$ cm; 95% CI: -1.78, 0.30; p-value for heterogeneity = 0.000; $I^2 = 87.8\%$, Fig. 6) with the random-effect model. Subgroup analysis was not conducted for birth chest circumference due to the limitation of studies in per subgroup. The result of Egger's linear regression test suggested no evidence of publication bias ($t = -0.24$, $P = 0.851$).

4. Discussion

Our overall meta-analysis presented a result of the reduction of newborns' birth weight, length and head circumference with maternal high-level arsenic exposure. However, there was no significantly pooled effect on the chest circumference of children, which may be related to the small number of included studies for this aspect. These finding potentially suggested that exposure to arsenic in life and work should be prevented in women of reproductive age.

The growth of a baby in intrauterine depends on multiple factors, including maternal nutrition and micronutrient intake during pregnancy, gestational age and maternal exposure to heavy metals in the environment (Kramer, 1987). During pregnancy, the fetus is entirely dependent on maternal sources of nutrition, and a balanced supply of maternal nutrients and micronutrients has a critical role on infant outcomes. Previous studies about the association between pregnancy dietary patterns and birth outcomes suggested that foods high in

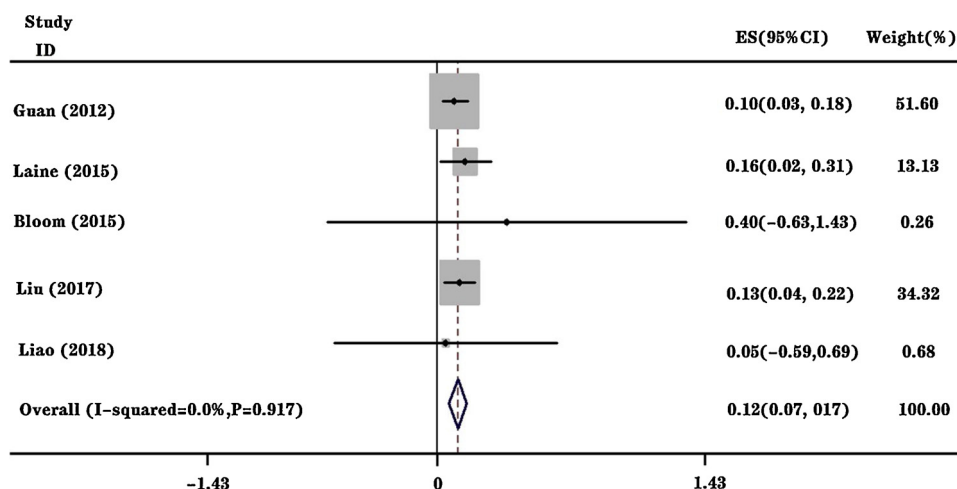


Fig. 4. Forest plot for the relation between arsenic exposure and birth length.

saturated and trans fats, refined sugar, or sodium are negatively associated with birth size, and that nutrient-rich foods such as fruits, vegetables, and whole grains were positively associated with birth size (Colon-Ramos et al., 2015; Grieger et al., 2014). Meanwhile, Harvey et al reported that maternal vitamin D supplementation significantly increased birthweight in infants (Harvey et al., 2014). The adverse effects of in utero exposure to environmental contaminants (polychlorinated biphenyls, methylmercury, lead, cadmium and arsenic) have opposing influences on fetal growth, because these pollutants can cross the placental and blood–brain barriers to reach the immature fetal organs and tissues (Miyashita et al., 2015). However, the association between maternal arsenic exposure and adverse birth size is not widely recognized, the results were also inconsistent.

Previous systematic review and meta-analysis (Quansah et al., 2015) has suggested the link of arsenic exposure and birth size, thereby demonstrating that arsenic exposure is an important risk factor for birth weight (-53.2 g, 95%CI: -94.9, -11.4). However, the meta-analysis was conducted in 2015 just included 4 studies to examine the relation between maternal arsenic exposure and birth weight. A recent review was conducted to discuss several studies investigating the association between maternal arsenic exposure and birth size (Milton et al., 2017). Of these included 11 studies, six support an increasing risk of low birth weight with greater exposure while four did not observed a statistically significant association. This review further revealed the association between maternal arsenic exposure and birth size. In our present study, we observed 25 g reduction in birth weight, 12 cm reduction in birth

length, 12 cm reduction in birth head circumference, and the association persisted in subgroup applying different biomarkers, and high quality studies. From the public health point of view, which provided convincing evidence to stimulate appropriate policy and mitigation measures for reducing maternal arsenic exposure.

Several mechanisms may explain the association between higher arsenic exposure and birth outcomes. Arsenic is a cause of maternal placental insufficiency and fetal intrauterine growth retardation, through the mechanism of stimulating the production of free oxygen radicals in the body and the disorder of the oxidative defence system. Moreover, reactive oxygen species could pass through the placenta and further affecting the fetal size (Chattopadhyay et al., 2002). Meanwhile, women of childbearing age have a more efficient methylation process compared to other times of their life due to the estrogen stimulated production of methyl groups required for sequential methylation of arsenic (Vahter, 2009). Animal studies had shown that arsenic is an endocrine disruptor that can hormone-activated gene transcription mediated, which affected the system of body's insulin growth factor, cell growth and glucose homeostasis even at low doses (Davey et al., 2007).

Subgroup analyses for maternal arsenic exposure and birth size except for the head circumference ($n = 3$) were conducted. The pooled regression coefficient of maternal arsenic exposure for newborn birth weight, length were higher in population applying the marker of exposure from human body. The possible reason maybe due to the marker of arsenic in the human body was more accurate and meaningful, and

Table 3

Subgroup analysis of arsenic and birth length.

Stratification group	n	β (95%CI)	Heterogeneity test			Publication bias	
			Q	p	I^2 (%)	t	p
Total	5	-0.12(-0.17, -0.07)	0.95	0.917	0.0	0.96	0.512
Location							
China	3	-0.11(-0.06, -0.17)	0.29	0.863	0.0	/	/
Americas	2	-0.17(-0.31, -0.02)	6.06	0.048	67.0	/	/
Combined	5	-0.12(-0.17, -0.07)	0.95	0.917	0.0	0.96	0.512
Exposure marker							
Urine	4	-0.14(-0.21, -0.06)	0.44	0.931	0.0	/	/
Blood	1	-0.10(-0.17, -0.03)	/	/	/	/	/
Combined	5	-0.12(-0.17, -0.07)	0.95	0.917	0.0	0.96	0.512
Study design							
PCO	4	-0.14(-0.21, -0.06)	0.44	0.931	0.0	/	/
CS	1	-0.10(-0.17, -0.03)	/	/	/	/	/
Combined	5	-0.12(-0.17, -0.07)	0.95	0.917	0.0	0.96	0.512

Abbreviations: CS, cross-sectional study; PCO, prospective cohort study; /, no value.

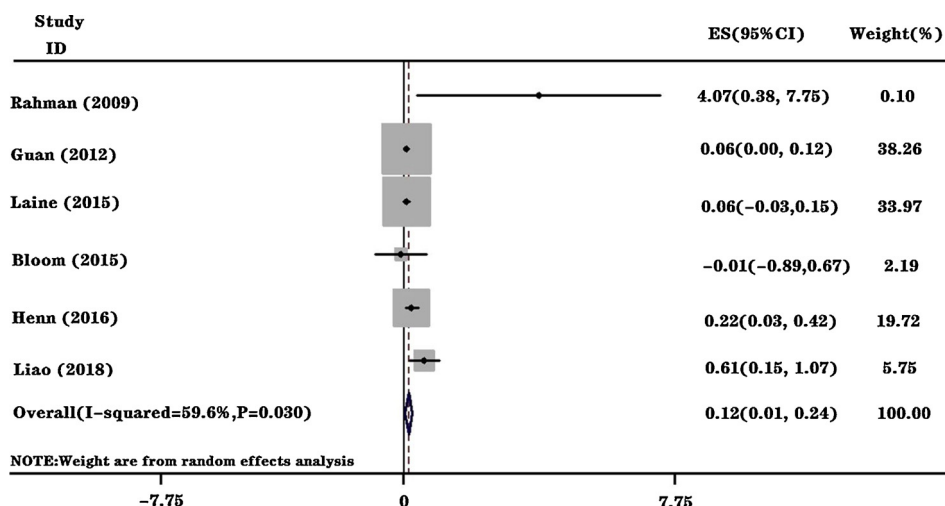


Fig. 5. Forest plot for the relation between arsenic exposure and head circumference.

the marker in the environment water was influenced by many unknown or not measured factors. According to the marker of measurement, there were five studies measured the urine, four studies measured the water, one study measured the hair and two study measured the blood. The level of arsenic in urine or blood could reflect the recent exposure to arsenic, and long-term arsenic intake was mainly accumulated in the body's hair or fingers/toes nails (Quansah et al., 2015). So more studies about the chronic exposure of arsenic for birth outcomes should be conducted. And that, human exposure to arsenic can via food, such as the fish, grains and vegetables. One study reported that both rice consumption and arsenic exposure through water were significantly associated with the higher concentration of total urinary arsenic (Gilbert-Diamond et al., 2011). The relationship between biomarkers (urine, blood and hair) and arsenic measured in diet or water mainly depend on the dose, frequency, duration of exposure, and metabolism of human. Further studies about maternal arsenic exposure and pregnancy outcomes should include food source of arsenic.

The marked heterogeneity in our meta-analysis refers to the difference of included studies from each other in the selection of study participants, study design, marker of exposure and outcomes assessment. We further conducted a subgroup analysis to explore the potentially main factors causing heterogeneity and influencing factors of results. Strong heterogeneity of our study was mainly attributed to the studies conducted in China, applying water as the marker and cross-

sectional study. The original studies also applied different exposure assessment methods and incorporated different exposure contrasts, which also may be the potential sources of the observed heterogeneity (Abhyankar et al., 2012). This approach could lead to misclassification of exposure, but the assumption is need to pool studies with different exposure metrics on the quantitative level. The New Hampshire Skin Cancer Study used different marker of exposure to examined the association between low-level arsenic exposure among US population and the risk of death from cardiovascular disease (Farzan et al., 2015). In the meantime, the subgroup analysis with varying categories in our study showed a similar conclusion. There was no publication bias in our study about the maternal arsenic exposure with birth size through funnel plots and egger's tests, which may be limited when the number of studies are small.

Our study has a number of strengths. Firstly, this study is more comprehensive and specific on the association between maternal arsenic exposure and birth size. Secondly, we conducted a subgroup analysis for these included studies on arsenic exposure and birth size according to the location of study, the marker of measurement and study design. Thirdly, we followed systematically, the guideline of the Preferred Reporting Items for Systematic Reviews and Meta-analysis. Most of the included studies were prospective cohort study with high quality.

Meanwhile, Several limitations also existed in our meta-analysis.

Table 4

Subgroup analysis of arsenic and birth head circumference.

Stratification group	n	β (95%CI)	Heterogeneity test			Publication bias	
			Q	p	I^2 (%)	t	p
Total	6	-0.12(-0.24, -0.01)	12.39	0.030	59.6	-0.41	0.719
Location							
China	2	-0.29(-0.82, 0.25)	5.40	0.02	81.5	/	/
Bangladesh	1	-4.07(-0.38, -7.75)	/	/	/	/	/
Americas	3	-0.10(-0.21, 0.01)	2.35	0.308	15.0	/	/
Combined	6	-0.12(-0.24, -0.01)	12.39	0.030	59.6	-0.41	0.719
Exposure marker							
Urine	4	-0.28(-0.78, 0.23)	10.00	0.019	70.0	/	/
Blood	2	-0.11(-0.26, 0.04)	2.37	0.124	57.8	/	/
Combined	6	-0.12(-0.24, -0.01)	12.39	0.030	59.6	-0.41	0.719
Study design							
PCO	5	-0.21(-0.46, -0.03)	11.62	0.020	65.6	0.45	0.730
CS	1	-0.06 (-0.12, -0.00)	/	/	/	/	/
Combined	6	-0.12(-0.24, -0.01)	12.39	0.030	59.6	-0.41	0.719

Abbreviations: CS, cross-sectional study; PCO, prospective cohort study; /, no value.

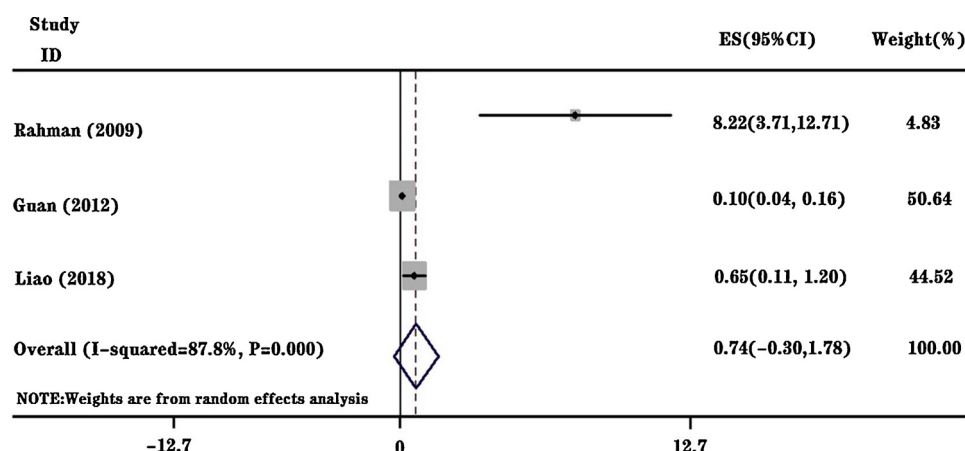


Fig. 6. Forest plot for the relation between arsenic exposure and chest circumference.

First, only published English literatures were included for analysis in our study, and did not investigate the existence of unpublished reports. Second, the results of subgroup analysis may be biased due to the small number of available studies. Finally, there were additional unidentified confounders (e.g. maternal nutrition, prepregnancy weight and occupational exposures) influenced birth size may exist across all included studies. There is unfortunately no information available on these variables for individual study subjects and they could not be adjusted for directly in the analysis.

5. Conclusions

In summary, maternal exposure to environmental arsenic was associated with the reduction of birth size. In addition, the reduction of effect for birth weight was stronger in women applying the marker of exposure in human body. The present results mostly pooled from studies conducted in the developing countries. Millions of people are still exposed to arsenic, and pregnant women eat arsenic-contaminated water and food. The affected countries should take concerted actions and effective mitigation programs to reduce environmental arsenic exposure of women in reproductive age and their children. Additional high-quality epidemiologic studies are necessary to assess the precise relationship between maternal arsenic exposure with birth size.

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Transparency document

The Transparency document associated with this article can be found in the online version.

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